School of Engineering

Cochin University of Science And Technology

B.Tech Degree 3rd semester 2nd series test in Mechanical Engineering Feb 2023

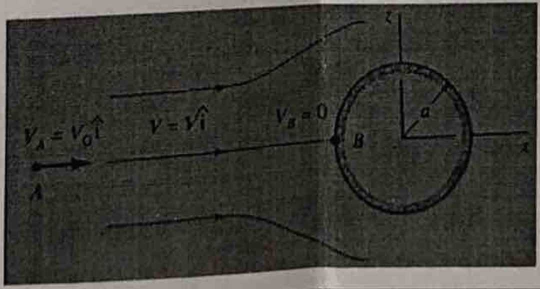
Time: 2 Hrs | Sub: ME19-205 1304 Fluid Mechanics | Max.Marks: 50

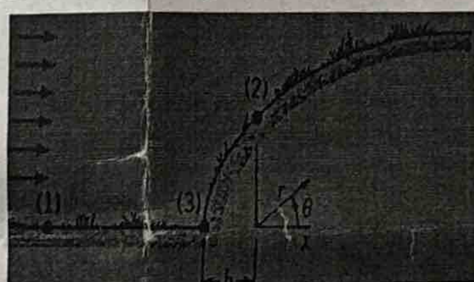
CO- Course Outcome, PO -Program Outcome, L- Blooms Taxonomy level.

CO2- Knowledge of Boundary layers and its separation

CO3- Apply the knowledge of buoyancy and floatation, hydrostatic forces on surfaces, flow rate and velocity measuring devices, notches and weirs, orifices, mouthpieces velocity boundary layer.

CO4- Apply the knowledge of flow field lines (path lines, stream lines etc), d Bernoulli's equation and potential flows and pipe flows.

1(a)	A rectangular pontoon is 5m long, 3m wide and 1.2 m height. The depth of immersion of the pontoon is 0.80 m in sea water. If the centre of the gravity is 0.6 m above the bottom of the pontoon, determine the meta-centric height. The density of sea water = 1023 kg/m^3. **(4)**	CO3	PO1 PO2	L2
1(b)	A pipe line which is 4 m in diameter contains gate valve. The pressure at the centre of the pipe is 20 N/cm^2. If the pipe is filled with oil of specific gravity 0.85, find the force exerted by the oil upon the gate and position of centre of pressure. **(4)**	CO3	PO1 PO2	L2
1(c)	A pitot-static tube is used to measure the velocity of water in a pipe. The stagnation pressure head is 7 m and static pressure head is 6 m. Calculate the velocity of flow assuming coefficient of tube equal to 0.97. **(3)**	CO3	PO1 PO2	L1 L2
2(a)	What is a venturimeter? Derive an expression for the discharge through a venturimeter. **(4)**	CO3	PO1 PO2	L1
2(b)	Water discharge at the rate of 98 litres/s through a 120 mm diameter vertical sharp-edged orifice placed under a constant head of 10 meters. A point, on the jet, measured from the vena contracta of the jet has coordinates 4.5 meters horizontal and 0.54 meters vertical. Find the coefficient C_v, C_c and C_d of the orifice. **(4)**	CO3	PO1 PO2	L3
3(a)	Consider an inviscid, incompressible, steady flow along a horizontal stream line A - B in front of a sphere of radius 'a' as shown in figure. From a more advanced theory of flow past a sphere, fluid velocity along the stream line is $V = V_0\left(1 + \frac{a^3}{x^3}\right)$. Derive an expression for the pressure variation along the stream line from point A far in front of the sphere ($x_A = -\infty$, $V_A = V_0$) to point B on the sphere ($x_B = -a$, $V_B = 0$). Also plot the variation qualitatively. **(7)** 	CO4	PO1 PO2 P12	L3

3(b)	Water is pumped through a 600mm diameter pipe with a head loss of 27m. In order to reduce the power consumption, it is proposed to lay another main of appropriate diameter (but with the same f and length 'l' as the first pipe) in parallel to the existing pipe and reduce the head loss to 9.6m still carrying the same total discharge. Find the diameters of the two pipes. (5)	CO2	PO1 PO2	L2 L3
3(c)	An oil of specific gravity 0.9 and viscosity 0.06 Poise is flowing through a pipe of diameter 200mm at the rate of 60 lit/sec. Find (i)the head loss due to friction for a 500m length of the pipe(ii) the power required to maintain the flow. (5)	CO4	PO1 PO2 P12	L3
3(d)	The shape of a hill arising from a plain can be approximated with the top section of a Rankine half body. The height of the hill approaches 60 m (i) when a 64km/hr wind blows toward the hill, what is the magnitude of the air velocity at a point on the hill directly above the origin (point 2)? (ii) What is the elevation of point (2) above the plain surface and (iii) what is the difference in pressure between point (1) on the plain far from the hill and point (2). Assume atmospheric pressure and temperature as 1 bar and 27°C. (8) 	CO4	PO1 PO2 P12	L3
4(a)	What is meant by boundary layer? What do you mean by separation of boundary layer? What is the effect of pressure gradient on boundary layer separation? What are the different methods of preventing the separation of boundary layers? (6)	CO2	PO1 PO2 PO12	L1 L3

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